Time-Consistent Bond Credit-Risk Early Warning and RD-Agent Governance: A Reliability Study

English translation of the corresponding Chinese research report.

Abstract

This study examines the problem of bond credit-risk early warning in a real-world financial database. It constructs a time-consistent dataset with the issuer-quarter as the unit of observation, using only ratings, financial variables, and derived indicators available before the prediction time point to predict whether an issuer will experience a downgrade in its issuer credit rating within the following six months.

The study simultaneously focuses on predictive performance, cross-period stability, and the protocol reliability of autonomous data-science processes. The experimental system includes a human-controlled baseline, rolling time backtests, hidden-test diagnostics, unlabeled feature-drift analysis, and progressive RD-Agent governance experiments that gradually narrow from full-model generation to restricted model recipes.

The human-controlled Protocol v2.2 LightGBM model was the only candidate model to pass the full protocol audit and be permitted to access the original hidden-test labels. Its PR-AUC on the locked validation set was 0.2764, and its PR-AUC on the 2023Q1-2025Q3 hidden test set was 0.0866. The test-period event rate was only 0.3429%; therefore, the test PR-AUC was 25.27 times the event-rate baseline. At a Top-1% review capacity, the model captured 33.33% of rating-downgrade events. An issuer-level cluster bootstrap gave a 95% interval of [0.0581, 0.1262] for test PR-AUC.

During rolling backtesting, the fixed model achieved PR-AUC values of 0.3646 and 0.2775 in 2019-2020 and 2021-2022, respectively. TRAIN-to-TEST drift analysis showed marked long-term distributional changes in cash, liquidity, rating-agency composition, and rating duration. The drift provides a reasonable data-level explanation for performance degradation but cannot by itself establish causal attribution.

Early RD-Agent experiments successively encountered problems in which validation labels participated in calibration, non-TRAIN data participated in constructing categorical vocabularies, and candidate selection was inconsistent. In controlled pseudo-prospective experiments, Phases 3A, 3B, and 3C respectively strengthened data isolation, structural constraints, and controller mediation, but all candidates were still blocked by the independent admission mechanism because of problems with categorical vocabularies, model structure, runtime interfaces, or machine-readable contracts for model recipes. By Phase 3C, no candidate had passed the then-preregistered scoring-admission protocol. Therefore, this study did not compute pseudo-test scores for those candidates or bypass the stopping rule through manual correction or additional attempts. Subsequent Phases 3D and 3E respectively evaluated controller recoverability and the first-attempt structural-compliance rate; the study endpoints for these phases still did not include pseudo-test performance scores.

Subsequent extension experiments further distinguished controller recoverability from agent-independent reliability. Without using performance feedback, Phase 3D used a frozen controller to normalize a near-compliant RD-Agent recipe into a structurally compliant candidate. Phase 3E then conducted repeated blind tests: among 15 primary evaluation samples, 3 first-attempt raw candidates passed the structural-compliance threshold, for a compliance rate of 20.0% and a Wilson 95% confidence interval of 7.05%-45.19%. This quantity measures interface and engineering compliance and does not represent predictive accuracy.

The results show that the reliability of real-world financial prediction depends not only on a model's ranking ability but also on point-in-time data engineering, cross-period evaluation, uncertainty analysis, and executable experimental governance. RD-Agent can be used to assist feature ideation and experiment generation, but in high-risk tasks it requires joint constraints from data boundaries, machine-readable contracts, immutable candidate freezing, independent protocol checks, and scoring-access controls.

Keywords: bond credit risk; rating downgrade; RD-Agent; time consistency; class imbalance; distribution drift; rolling backtesting; model governance

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1. Research Background and Business Issues

The credit profile of a bond issuer directly affects funding costs, credit spreads, and portfolio risk. Compared with default prediction, rating-downgrade events occur more frequently and typically provide an earlier indication of deteriorating credit quality, making them a suitable intermediate-risk outcome for credit-risk early-warning systems.

The core issues of this project are:

> At the quarter-end time point t, predict whether an issuer will experience a downgrade of the entity's credit rating within the next six months using only information that has been public or available before that point in time.

This task has the following characteristics:

- The proportion of positive examples is very low, which is a typical rare event classification problem;
- Financial data are subject to disclosure lags and statement revisions;
- The same issuer reappears in multiple quarters;
- Random partitioning can cause time leakage and overestimate performance;
- There may be significant distribution drift between the validation set and future periods;
- Model outputs must be interpretable and auditable by business personnel.

Accordingly, this study does not treat predictive performance as its sole objective; it also examines temporal consistency, reproducibility, probability quality, and automated-experiment governance.

2. Data Sources and Data Engineering

2.1 Data Source

The research data are derived from publicly available market data and are accessed through a read-only Oracle database environment. They mainly include:

- Entity ratings and rating change records;
- Bond credit event records;
- Balance-sheet data;
- Income-statement data;
- Cash-flow-statement data;
- Data dictionary and ETL mapping information.

All database operations are read-only queries. Subsequent cleaning, feature construction, and modeling are performed in a locally isolated environment.

2.2 Prediction Units and Labels

Prediction units are defined as issuer-quarter snapshots. Each row represents an issuer's observable state at quarter end.

The label `DOWNGRADE_6M` is defined as follows:

$$Y_{i,t} = 1, \quad \text{if issuer i experiences a credit-rating downgrade within } (t,\ t + 6 \text{ months}];$$
$$Y_{i,t} = 0, \quad \text{otherwise.}$$

This definition clearly separates the prediction time point from the future observation window and avoids using information after the prediction time point as model features.

2.3 Point-in-Time Dataset

The financial statements are not directly linked by reporting period, but are time-consistently matched according to the actual announcement date. For each quarterly snapshot, only the latest available records with an announcement date no later than the snapshot date are selected.

The main audit contents include:

- Whether the financial announcement date is later than the snapshot date;
- How to choose between multiple versions of the same reporting period;
- Quarterly financial-data coverage rates;
- Missingness rates of original fields and derived features;
- Whether the label window and the feature window overlap;
- Whether identifier and time-control fields enter the model.

The final modeling data are divided by time as follows:

Split	Period	Issuer-quarter observations
TRAIN	2013Q2-2020Q4	172,566
VALID	2021Q1-2022Q4	71,194
TEST	2023Q1-2025Q3	114,603

3. Experimental Protocol and Evaluation Methods

3.1 Protocol v2.2

To control temporal leakage and deviations in automated experiments, this study established Protocol v2.2, which mainly stipulates the following:

- Only TRAIN is used to fit the model and all learned preprocessing;
- VALID is used only for early stopping, model comparison, and hyperparameter selection;
- TEST does not participate in preprocessing, feature engineering, calibration, or candidate selection;
- `ROW_ID`, `SPLIT`, time fields, and institution identifiers are prohibited as predictive features;
- Reconstruction of issuer or time information through `ROW_ID` is prohibited;
- Submitted results must come from the candidate model with the highest PR-AUC on the locked validation set;
- Only candidate models that pass independent protocol audits are granted access to hidden-test labels.

3.2 Evaluation Metrics

Because the proportion of positive examples is very low, the primary metric is PR-AUC, or Average Precision (AP). Compared with ROC-AUC, PR-AUC is more sensitive to performance in identifying the minority class.

Also reported:

- Event rate;
- Improvement factor relative to the event rate;
- Brier score;
- The difference between the average predicted probability and the actual occurrence rate;
- Bootstrap confidence interval;
- Quarterly stability.

3.3 Hidden Test-Set Governance

Hidden-test labels are isolated from the data directory accessible to the agent. RD-Agent candidate models with invalid protocols must not undergo hidden-test scoring even if `submission.csv` is successfully generated.

Therefore, the absence of hidden-test scores does not indicate a failed run; it indicates that the experimental-governance mechanism successfully prevented noncompliant models from reaching final evaluation.

4. Human-Controlled Baseline

The human-controlled candidate model uses LightGBM and strictly follows Protocol v2.2. Categorical vocabularies, missing-value rules, and other learned preprocessing are all fitted on TRAIN only; VALID is used only for early stopping and model selection.

The model's PR-AUC on the locked validation set is:

$$\text{PR-AUC}_{\text{VALID}} = 0.2764$$

This candidate model passed an independent protocol check; its code, result files, and hashes were then locked, after which one hidden-test evaluation was conducted. Its hidden-test PR-AUC is:

$$\text{PR-AUC}_{\text{TEST}} = 0.0866$$

There is a marked decline in performance between the validation and test sets, indicating that reliance on a single validation period may overestimate the model's ability to generalize to data from the more distant future.

5. RD-Agent Reliability Experiment

5.1 Experiment 1: Prompt-Based Protocol Constraints

The first official RD-Agent experiment provided Protocol v2.2 to the agent only through the task instructions.

The agent successfully completed modeling and generated a submission file, but the audit found the following:

- Fitted an isotonic regression calibration model using VALID labels;
- Derived time information from `ROW_ID`;
- Used the quarter field as a predictor variable;
- Used time weights;
- The final submitted model was not the candidate model with the highest performance on the locked validation set.

This experiment shows that the agent can interpret protocol text but cannot reliably implement every restriction in generated code.

5.2 Experiment 2: Removing Time and Identifier Fields

The second experiment directly removed time fields and issuer identifiers from the data accessible to the agent and replaced `ROW_ID` with an unparseable random identifier.

This measure successfully prevented direct time leakage and identifier reconstruction, but the agent still combined TRAIN and TEST to jointly construct a categorical vocabulary.

This shows that removing clearly leaky fields is insufficient: the test set can also affect the model through preprocessing statistics.

5.3 Experiment 3: TRAIN-Only Categorical Encoding

The third experiment further removed the original categorical field and used a categorical encoding fitted on TRAIN only. Unseen categories were uniformly mapped to a reserved code.

Under this structural constraint, no test-set involvement in preprocessing was found. The best validation PR-AUC of a single model generated by RD-Agent was 0.2841, exceeding the human-controlled model's 0.2764.

However, the agent ultimately submitted an ensemble model with a validation PR-AUC of 0.2816 rather than the higher-performing single model and therefore still violated the prespecified candidate-selection rule.

5.4 Experimental Comparison

Experiment	Constraint Mode	Best VALID PR-AUC	Submitted-Model VALID PR-AUC	Protocol Result
Human-controlled	Protocol v2.2	0.2764	0.2764	Pass
RD-Agent 04	Prompt-based protocol	0.2704	0.2636	Fail
RD-Agent 05	Remove time and identifier fields	0.1109	0.1109	Fail
RD-Agent 06	Add TRAIN-only categorical encoding	0.2841	0.2816	Fail

These results show that structural constraints do gradually eliminate different types of data leaks, but structural constraints alone cannot guarantee correct candidate selection and submission governance.

6. Rolling Time Backtesting

To examine the model's stability across periods, this study fixed the human-controlled LightGBM specification and performed expanding-window backtests.

6.1 Fold 1

- Training period: through 2018Q4;
- Validation period: 2019Q1-2020Q4;

- Validation samples: 60,702;
- Positive examples: 652;
- Event rate: 1.0741%;
- PR-AUC: 0.3646;
- Relative improvement over the event rate: 33.94-fold.
- Brier score: 0.00920.

6.2 Fold 2

- Training period: through 2020Q4;
- Validation period: 2021Q1-2022Q4;
- Validation samples: 71,194;
- Positive examples: 521;
- Event rate: 0.7318%;
- PR-AUC: 0.2775;
- Relative improvement over the event rate: 37.93-fold.
- Brier score: 0.00712.

Fold 2 differs from the locked human-controlled validation result by only about 0.00114, indicating that the backtest implementation is broadly consistent with the formal benchmark.

Because Fold 3 corresponds to 2023-2024 and overlaps with the hidden-test period, this study did not access its labels during the development phase.

7. Uncertainty and Quarterly Stability

The conditional uncertainty of PR-AUC was estimated using 2,000 stratified observation-level bootstrap iterations:

- Fold 1: 0.3646, with a 95% interval of [0.3292, 0.4007];
- Fold 2: 0.2775, with a 95% interval of [0.2401, 0.3205].

The two intervals do not overlap, providing relatively strong descriptive evidence of weaker ranking performance in the later period. However, this bootstrap does not preserve within-issuer correlation and keeps the numbers of positive and negative samples fixed, so it is not interpreted as a formal paired significance test.

Quarterly PR-AUC fluctuates from 0.0468 to 0.6689. There were relatively few positive examples in 2022Q2 and 2022Q3, and the single-quarter PR-AUC was particularly unstable.

This suggests that credit-risk models should not report only an overall score; they should also continuously monitor:

- Quarterly event rate;
- PR-AUC and lift;
- Mean predicted probability;
- Calibration error;
- Feature missingness rates;

- Rating-agency and rating-grade distributions; and
- Stability of high-risk lists.

8. Comprehensive Diagnostic Evaluation of the Hidden Test Set

8.1 Evaluation Scope and Method Boundaries

The human-controlled Protocol v2.2 LightGBM candidate model had its code, features, prediction files, and SHA-256 locks completed before it accessed hidden-test labels, and it passed an independent protocol audit. This section provides supplementary diagnostics of this candidate's fixed predictions after one formal hidden-test evaluation.

This analysis is a **post hoc descriptive evaluation of the locked model**. Test labels were not used for model fitting, feature selection, hyperparameter selection, calibration, or comparison of candidate models. The results in this section must not be used to select subsequent RD-Agent candidates or tune new models. The parsing of `ROW_ID` in the evaluation script is used only to recover quarter and issuer-cluster identifiers and is not used as input to any prediction model.

The test set contains 114,603 issuer-quarter observations, including 393 rating downgrade events, with an event rate of 0.3429%.

8.2 Overall Ranking and Probability Error

Model	PR-AUC	Event rate	PR-AUC / event rate	ROC-AUC	Brier score	Mean predicted risk
Human-controlled LightGBM	0.0866	0.3429%	25.27 times	0.8735	0.00439	0.7698%
Weighted logistic regression baseline	0.0527	0.3429%	15.37 times	0.8685	0.07593	17.6929%

The human-controlled model's PR-AUC on the test set is 0.0866. Because the event rate in this period is only 0.3429%, its PR-AUC is 25.27 times the random-ranking baseline. Therefore, the numerical value of 0.0866 cannot be interpreted in isolation from this extremely low event rate.

The weighted logistic-regression baseline had PR-AUC of 0.0527, or 15.37 times the event rate. The human-controlled model's PR-AUC was 0.0339 higher.

The human-controlled model's mean predicted risk is 0.7698%, above the observed event rate of 0.3429%, and its Brier score is 0.00439. The mean predicted risk of weighted logistic regression reaches 17.6929%, indicating that the raw output after class weighting cannot be directly interpreted as the actual event probability. The baseline's higher Brier score mainly reflects its severely upward-biased probability scale, rather than merely a difference in ranking performance.

8.3 Event Capture Under Fixed Audit Capacity

Review capacity	Number of observations reviewed	Human-controlled model			Baseline	
		Capture count	Recall	Precision	Capture count	Recall
Top 1%	1147	131	33.33%	11.42%	106	26.97%
Top 5%	5731	231	58.78%	4.03%	218	55.47%
Top 10%	11461	265	67.43%	2.31%	259	65.90%

If quarterly risk-screening capacity is approximately represented by a priority review list of a fixed proportion, the human-controlled model captured 131 downgrade events in the Top 1% during the full test period, accounting for 33.33% of all 393 events; it captured 231 events in the Top 5%, with a recall of 58.78%, and 265 events in the Top 10%, with a recall of 67.43%.

Precision for the top 1% list is 11.42%, approximately 33.31 times the overall event rate. This result suggests that the model is better suited to relative risk ranking and analyst work-queue generation than to interpreting raw scores that have not undergone external calibration as absolute downgrade probabilities.

8.4 Issuer-Level Cluster Bootstrap

Because the same issuer appears repeatedly across multiple quarters, 2,000 Cluster Bootstrap iterations are performed in this section, using issuers as the resampling unit. For each resampling iteration, the fixed model predictions are kept unchanged, and PR-AUC is recalculated for the human-controlled model, the logistic-regression baseline, and the difference between the two.

The issuer-level bootstrap 95% interval for the human-controlled model's PR-AUC is [0.0581, 0.1262], versus [0.0362, 0.0768] for the logistic-regression baseline. The 95% interval for the paired PR-AUC difference is [0.0147, 0.0618].

In all 2,000 resampling iterations, the PR-AUC of the human-controlled model was higher than that of the logistic-regression baseline. This result shows that, under fixed test predictions and issuer-cluster resampling, the observed difference between the models has relatively strong stability. However, this bootstrap did not retrain the models or repeat the execution of feature engineering and model selection; it therefore reflects only the conditional uncertainty of fixed candidates under changes in test-sample composition and should not be interpreted as an unconditional confidence interval or a formal significance test for the entire research-and-development process.

8.5 Quarterly Event Distribution During the Test Period

Quarter	Number of observations	Number of downgrade events	Event rate	PR-AUC	Mean predicted risk
2023Q1	9460	40	0.423%	0.1770	1.287%
2023Q2	9642	8	0.083%	0.0586	0.822%
2023Q3	9785	11	0.112%	0.0624	0.522%
2023Q4	9885	99	1.002%	0.1069	0.849%
2024Q1	10055	102	1.014%	0.1006	1.062%
2024Q2	10626	9	0.085%	0.1969	0.745%
2024Q3	10688	7	0.065%	0.0913	0.488%
2024Q4	10803	45	0.417%	0.1103	0.765%
2025Q1	11010	50	0.454%	0.1328	1.002%
2025Q2	11231	12	0.107%	0.0804	0.685%
2025Q3	11418	10	0.088%	0.1452	0.335%

Quarterly event counts during the test period were highly uneven. There were 99 and 102 downgrade events in 2023Q4 and 2024Q1, respectively, but only 8, 9, and 7 events in 2023Q2, 2024Q2, and 2024Q3. Single-quarter PR-AUC is therefore simultaneously affected by ranking quality, event count, and market conditions.

Overall test PR-AUC does not indicate that the model performs the same in every quarter. Subsequent drift analysis needs to further distinguish between event rate drift, input feature drift,

rating agency composition changes, and feature-label relationship changes.

8.6 Section Conclusion

The comprehensive diagnostics show that the human-controlled model's PR-AUC declined from 0.2764 on the validation set to 0.0866 on the test set, indicating substantial performance decay across periods; however, the test-period event rate was only 0.3429%, and the model still achieved a PR-AUC 25.27 times the event-rate baseline and captured one-third of downgrade events under a Top-1% review capacity.

Therefore, the test results do not support the explanation that the model has lost all predictive value, nor do they support claims of stable generalization based solely on the results of the validation period. A more reasonable conclusion is that the model still retains meaningful risk-ranking signals in test data from a more distant future period, but performance and probability scales are affected by significant temporal changes, requiring further study through drift diagnostics, rolling validation, and new future evaluation windows.

9. Feature-Distribution Drift and Performance-Degradation Diagnosis

9.1 Analysis Purpose and Method Boundaries

The human-controlled model's PR-AUC declined from 0.2764 on the validation set to 0.0866 on the hidden test set, indicating that the model exhibited marked degradation in ranking performance in a more distant future period. To determine whether this phenomenon was accompanied by changes in input data distributions, the study conducts Population Stability Index (PSI) analysis of TRAIN, VALID, and TEST modeling features without accessing hidden-test labels.

This analysis uses only feature values, missingness status, and data-split information. It does not use the `DOWNGRADE_6M` label or perform model fitting, hyperparameter tuning, feature filtering, or comparison of candidate models. This section is therefore an unlabeled data diagnostic conducted after model freezing and cannot be used to retrospectively optimize candidate models that have completed hidden-test evaluation.

For feature X_j, the reference-period data are divided into B intervals or categories. The Population Stability Index is defined as:

$$PSI_j = \Sigma_{b=1}^{B} (q_{jb} - p_{jb}) \ln[(q_{jb} + \varepsilon) / (p_{jb} + \varepsilon)]$$

Here, p_{jb} and q_{jb} denote, respectively, the proportions of samples falling into bucket b during the reference and comparison periods, and ε is a smoothing term that prevents numerical divergence caused by zero probabilities. Numeric-feature bucket boundaries are determined only from the reference period, and missing values are retained as separate buckets. Categorical features likewise handle missing categories separately from unknown categories appearing in the comparison period.

This study uses PSI values of 0.10 and 0.25 as descriptive monitoring thresholds: values below 0.10 indicate a relatively stable distribution, values from 0.10 to 0.25 indicate that attention is needed, and values of at least 0.25 indicate substantial distributional change. These thresholds are used for screening and ranking rather than as tests of statistical significance.

9.2 Data Structure and Pre-Checking

The drift analysis covers 97 protocol-compliant modeling features, including 90 numerical and 7 categorical features. TRAIN and VALID together contain 243,760 issuer-quarter observations, and TEST contains 114,603 observations. There were 9,557 issuers in the training and development period and 11,627 issuers in the test period. Of these, 9,534 issuers appeared in both periods, and the test period included 2,093 new issuers.

The full audit field list contains 16 more fields than the final modeling export. These fields are all `IDENTIFIER_NOT_FEATURE` or `AUDIT_NOT_FEATURE` fields, such as issuer identifiers, snapshot dates, financial-report dates, actual announcement dates, and data-source record identifiers, and have been physically removed from the modeling export. `ROW_ID` is a control key newly added during the export stage and is permitted only for external row alignment, issuer clustering, and temporal diagnosis and must not enter the model after parsing.

Preliminary checks showed that 28 features had absolute changes of at least 5 percentage points in missingness rates between TRAIN and TEST. Three categorical features had categories in TEST that were unseen during TRAIN, but the largest unknown-category share was only 0.4424%, indicating that unknown categories were not the main source of overall drift.

9.3 Overall Drift Results

Comparison period	PSI ≥ 0.25	0.10 ≤ PSI < 0.25	Main interpretation
TRAIN → TEST	9	22	Widespread distributional changes exist over long time spans
VALID → TEST	1	2	Drift is more concentrated in the recent period, but substantial local changes remain.

The drift from TRAIN to TEST is markedly stronger than from VALID to TEST. This difference indicates that some distributional changes are long-term structural changes formed through accumulation over many years, rather than all having occurred before and after the test period.

Representative features with the highest PSI in TRAIN-TEST include:

Representative feature	PSI	Main diagnosis
Year-on-year growth in cash deposited with banks	1.776	Changes in the observed-value distribution and missingness status occur simultaneously
Year-on-year change in the cash ratio	1.679	Changes in the observed-value distribution and missingness status occur simultaneously
Level of cash deposited with banks	1.104	Both the long-term coverage rate and the distribution of values change

Representative feature	PSI	Main diagnosis
Cash ratio	1.078	The distribution of liquidity features changes markedly
Rating agency	0.393	The composition of existing rating agencies changes
Rating duration	0.312	The proportion of samples with a long rating duration increases
Year-on-year change in rating duration	0.268	Rating-dynamics features change

In VALID–TEST, the most pronounced change is concentrated in the year-on-year change in rating duration, whose PSI is 0.4466; the PSI for rating agency is 0.1235. The proportion of samples for which the year-on-year change in rating duration exceeds 365 days increased from approximately 1.60% to approximately 18.00%, constituting the main source of recent drift.

To avoid repeatedly interpreting multiple encodings of the same business quantity as multiple independent drift mechanisms, the rating-agency name and code, the raw-value and signed-log representations of cash deposited with banks, and equivalent expressions related to leverage and equity were combined during aggregation. The formal results file retains all field-level calculations; the report figures display only representative variables with distinct business meanings.

9.4 PSI Contribution Breakdown

A single overall PSI value cannot distinguish whether drift arises from changes in valid observed values, changes in missingness rates, or an increase in unknown categories. Therefore, this study further decomposes the PSI of each feature as follows:

$$\mathrm{PSI} = \mathrm{PSI}_{\mathrm{observed}} + \mathrm{PSI}_{\mathrm{missing}} + \mathrm{PSI}_{\mathrm{unseen}}$$

These three components respectively represent the contributions of existing observed values or categories, missing-value buckets, and unknown-category buckets.

The contribution audit covers a total of 45 "feature-period comparison" combinations. The results show:

- Five combinations can be classified as dominated by changes in missingness;
- No combinations are dominated by unknown categories;
- No high-drift feature with a PSI of at least 0.25 can be fully explained by changes in missingness rates.

For cash deposited with banks, cash ratios, and their year-on-year variables, missing-bucket contributions account for approximately 31%–46% of their respective PSI values, showing that changes in financial-data coverage are indeed important; however, the remaining contribution still comes from the redistribution of non-missing observed values. Therefore, these high-PSI results cannot simply be attributed to more complete data in later periods.

Rating-agency drift mainly arises from changes in the composition of existing rating agencies, rather than from the addition of unknown rating agencies. The sample proportion of one existing rating agency increased from about 0.11% to about 4.59%, making it an important contributor to the long-term drift of this variable. For rating duration, the proportion of samples exceeding 1,351 days increased from approximately 10.0% to approximately 29.5%, indicating a marked difference

between the rating-record structures of issuers in the test period and in the training period.

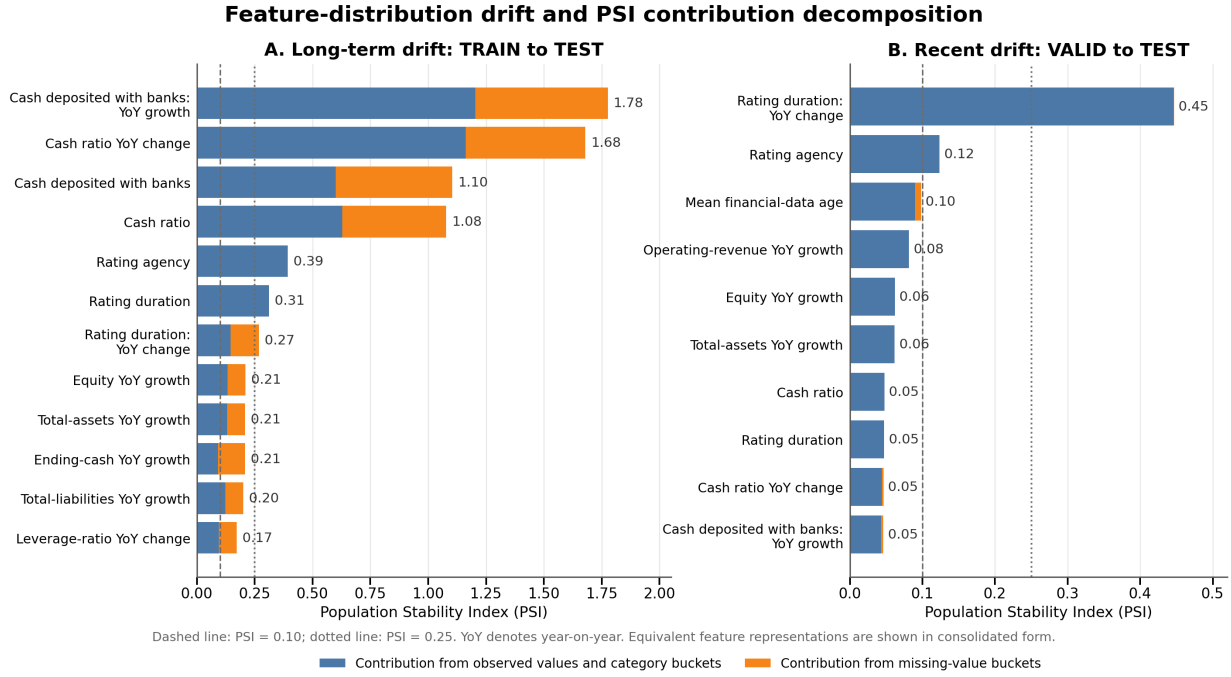


Figure: Feature distribution drift and PSI contribution decomposition. Drawn from this project's drift-audit results, the left panel compares TRAIN with TEST, representing cumulative change over multiple years; the right panel compares VALID with TEST, representing more recent change nearer the deployment period. Blue represents the contribution from observed values and category buckets, including any unseen-category contribution, and orange represents the contribution from missing-value buckets. Dashed and dotted lines mark the descriptive PSI monitoring thresholds of 0.10 and 0.25, respectively.

9.5 Relationship to Performance Degradation in the Test Period

Drift analysis provides a reasonable data-level explanation for performance degradation from the validation set to the test set: variables on which the model relies, such as liquidity, cash, rating-agency composition, and rating duration, show distributional changes to varying degrees during the test period. In particular, long-term drift from TRAIN to TEST is far stronger than that from VALID to TEST. This suggests that a mapping established using a single historical training window may gradually deviate from the data-generating mechanism in subsequent periods.

However, PSI measures only marginal feature-distribution changes:

$$P_t(X_j) \neq P_{t'}(X_j)$$

It cannot directly prove that the conditional relationship has also changed:

$$P_t(Y \mid X) \neq P_{t'}(Y \mid X)$$

Nor can it prove that a particular high-PSI feature causes a decline in PR-AUC. Performance changes may also arise from changes in event rates, rating policies, issuer composition, and feature interactions, as well as drift in feature-label relationships. Therefore, the conclusion of this section is that performance degradation accompanies observable input drift, rather than establishing a causal attribution for the decline in performance.

Because this section does not use TEST labels at all, the drift results can serve as a basis for deployment monitoring and future retraining trigger mechanisms. They must not be used to reselect a model that has already received one hidden-test evaluation. Methods for subsequent model improvements should be prespecified in a new development window, and those improvements should be evaluated using a new future-label window.

9.6 Section Conclusion

Feature-drift analysis shows that the test period is not a static extension of the training-period data. Long-term changes are widespread among features such as cash, liquidity, rating agencies, and rating duration; recent changes are more concentrated in rating-duration dynamics and rating-agency composition. Contribution decomposition further shows that high PSI values often reflect both changes in valid observed-value distributions and changes in data coverage and cannot be explained solely by changes in missingness rates.

This result supports the integration of drift monitoring into the formal governance processes of credit-risk early-warning systems. After the model is deployed, monitoring should include not only overall PR-AUC and the event rate, but also the PSI of representative features, missing-bucket contributions, the proportion of unknown categories, rating-agency composition, and rating-duration distributions. When drift crosses a preset threshold, data audits, rolling backtesting, and retraining evaluation should be triggered, rather than directly and automatically replacing the existing model.

10. Controlled Pseudo-Prospective Reliability Experiment for RD-Agent

10.1 Experimental Purpose

The preceding experiments show that prompt constraints, field removal, and TRAIN-only categorical encoding can progressively reduce data leakage, but still cannot guarantee that RD-Agent complies with candidate-selection, model-structure, and submission rules. To further examine the reliability of autonomous agents under machine-executable constraints, this study designed a retrospective pseudo-prospective experiment.

This experiment does not aim to repeatedly search for higher-scoring models. It examines the following question: when the training period, model interface, number of candidates, independent auditing, and scoring permissions are all locked in advance, can RD-Agent generate a candidate model eligible for scoring?

The experiment ultimately comprises five progressive sub-experiments, from Phase 3A to Phase 3E. The sub-experiments progressively identify problems involving data boundaries, code execution, machine contracts, controller recovery, and the independent first-attempt compliance rate. They should not be interpreted as independent comparisons of model performance.

10.2 Time Window and Data Isolation

The pseudo-prospective experiment reuses a historical period for which observation has been completed to avoid accessing data from the original hidden-test period, 2023–2025:

Data role	Time range	Number of observations	Number of positive examples	Event rate
TRAIN	2013Q2-2018Q4	111,864	1,444	1.2909%
Internal validation window	2019Q1-2020Q4	60,702	652	1.0741%

Data role	Time range	Number of observations	Number of positive examples	Event rate
Pseudo-test window	2021Q1-2022Q4	71,194	521	0.7318%

The agent can access only data packages consistent with the current experimental phase. Pseudo-test labels are kept in a private control directory, and candidates can enter the scoring stage only after passing through freezing, an independent protocol audit, and admission determination. Pseudo-test labels, raw time fields, and parsable issuer identifiers must not be used during the agent-generation process.

To reduce result-oriented adaptive attempts, the experiment also prespecifies the number of candidates and termination conditions. Code, logs, hashes, and audit conclusions are retained in the event of candidate noncompliance, but the admission mechanism must not be bypassed in order to obtain a score.

10.3 Phase 3A: Full-Model Candidates After Data Isolation

Phase 3A gives RD-Agent an isolated training and validation interface and requires it to generate full-model candidates. After the candidate run, the code and prediction files are locked and an independent checker performs a static protocol audit.

The audit found two mandatory failures:

- The categorical vocabulary still relied on non-TRAIN data, violating the TRAIN-only categorical-vocabulary constraint;
- The candidate used the average of predictions from multiple random seeds, violating the preregistered single-model, no-ensemble rules.

Additionally, the audit documented a non-mandatory warning: the candidate performed exploratory analysis of validation features. The final protocol status for Phase 3A is `BLOCKED`; private labels were not accessed, and the candidate received neither external-validation nor pseudo-test scores.

This result illustrates that even if labels are isolated, agents may still push the experimental boundaries through unlabeled test features or model combinations. Reliable governance requires not only isolating labels, but also limiting the sources of preprocessing statistics and allowed model structures.

10.4 Phase 3B: Rerun Under Structured-Data Constraints

Phase 3B further converted categorical variables into TRAIN-only representations and strengthened requirements including use of a single model, compatibility with any sample size, use of identifiers only for alignment, and prohibition of validation-feature exploration.

No scorable candidates were formed during the official run. The revised independent audit confirmed five mandatory failures:

- No complete, valid candidate output was produced;
- The code could not reliably handle any sample size permitted by the protocol;
- The candidate still included validation-feature exploration;
- Use of `ROW_ID` extended beyond pure alignment; and
- The use of prediction averaging violated the single-model constraint.

The audit also issued a warning about the risk of duplicated class weighting. Neither validation labels nor pseudo-test labels were accessed, and no performance scoring was performed.

Phase 3B demonstrates that physical removal of high-risk fields can narrow agent permissions, but cannot alone ensure the execution completeness and algorithmic-structure compliance of generated code. Field-level isolation, code-level auditing, and output-level admission checks must all remain in place.

10.5 Phase 3C: Controller-Mediated Restricted Recipe Generation

To further reduce agent permissions, Phase 3C no longer allows RD-Agent to write the full training and scoring process directly. The agent can submit only a model recipe constrained by a JSON Schema and a feature whitelist; implementation of feature operators, data reading, LightGBM training, prediction alignment, and scoring are all handled by the external controller.

The protocol prespecifies up to two opportunities for recipe generation. The second opportunity is used only to correct technical or interface issues, and feedback must not be based on external validation performance; any manual modification causes the candidate to lose its status as "RD-Agent Raw Output."

Three types of errors occurred in the first candidate:

- The number of inputs for an engineered feature was inconsistent with its operator definition;
- LightGBM parameters prohibited by the protocol were used;
- Required parameters such as `max_bin`, `max_depth`, and `min_child_samples` were missing.

Four types of contract errors occurred in the second candidate:

- An additional top-level field, `model_hyperparameters`, was used;
- The specified `lightgbm_params` object was not provided;
- All ten required LightGBM parameter fields were missing;
- Required top-level field `lightgbm_params` was missing.

The candidates from the two attempts were both rejected at the independent machine-inspection stage. Under the preregistered rules, a third attempt was not allowed, nor could a candidate whose fields had been manually renamed be treated as an RD-Agent candidate. Therefore, the controller did not carry out model training, and validation features, validation labels, pseudo-test features, and pseudo-test labels did not enter the scoring process.

The final status of Phase 3C is `CLOSED_NO_PROTOCOL_ELIGIBLE_RECIPE`.

10.6 Phase 3D: Schema-Guided Controller Recovery

Phase 3D does not test whether RD-Agent can independently produce a compliant result on a single attempt. Instead, it tests whether a near-compliant agent candidate can be recovered deterministically under pre-frozen controller rules. The agent provided a modeling recipe, 15 engineered-feature definitions, TRAIN-based class-weight selection, bounded LightGBM parameters, and recipe documentation; the controller performs only pre-specified interface normalization.

The first normalized candidate still failed independent inspection. The subsequently formed v2 candidate carried out three types of auditable corrections: exact parameter keys were normalized, obsolete documentation references were corrected, and the `epsilon` field was removed from six non-ratio operators that did not require it. The permitted `epsilon` in the ratio-safety operator remained unchanged, and the remaining contents of the candidate were not modified. The recovery

process used neither validation-set nor pseudo-test-set data and used no performance metrics.

The independent recipe Schema, whitelist, operator, prohibited-input, and LightGBM-parameter checks all eventually passed; inventory and output audits also both passed. The frozen candidate contained 230 permitted base features and 15 engineered features, and its output format satisfied the protocol requirements.

The Phase 3D conclusion is therefore strictly limited to the following: the frozen controller can deterministically recover an RD-Agent-generated near-compliant candidate into a recipe that complies with structural and data-boundary requirements. It does not demonstrate independent first-attempt success by RD-Agent or improved predictive performance.

10.7 Phase 3E: Independent Blind-Test Compliance Reliability

Phase 3E repeatedly examines RD-Agent's first raw `evo_loop_0` candidate under the same frozen task, Schema, feature whitelist, runtime image, and independent admission standards. The experiment prohibits controller repair, feedback-driven modification, performance-driven selection, and network access. Each candidate is replayed in a separate workspace and subjected to machine inspection.

There were 18 official launches. `run_001`, `run_002`, and `run_010` were excluded from the denominator because early implementation deviations or controller-environment binding errors made them ineligible for primary evaluation. The final sample comprised 15 primary evaluations, of which 3 passed and 12 failed:

- First-candidate structural-compliance rate: 3/15, or 20.0%;
- Wilson 95% confidence interval: 7.05%-45.19%;
- Successful runs: `run_005`, `run_007`, and `run_018`;
- Number of failed primary-evaluation samples: 12.

Failure types are as follows:

Failure type	Count	Share of all failed primary-evaluation samples
Raw-code syntax error	3	25.0%
Multiple Schema-contract violations	2	16.7%
Schema validation aborted before output	2	16.7%
Prohibited use of epsilon	1	8.3%
LightGBM parameter contract violation	1	8.3%
Output directory contract violation	1	8.3%
Recipe-version contract violation	1	8.3%
Schema-constant mismatch	1	8.3%

These failures occurred primarily at the software-contract level, involving syntax, Schema, parameter names, versions, value ranges, and output location. They reflect software-contract issues rather than a demonstrated lack of predictive capability. Likewise, Phase 3E success means only that a first-attempt candidate met the structural-compliance endpoint; it does not show that its model performance exceeded that of the human-controlled benchmark.

The experiment followed prespecified administrative time and capacity criteria and stopped after 15 primary evaluation samples. The stopping decision did not depend on the outcome of `run_018` or any other run. Because the sample size is limited, confidence intervals remain wide and the findings apply only to this task, model configuration, temperature setting, and locked environment.

10.8 Why Are Pseudo-Test Scores for RD-Agent Not Reported?

This study does not report pseudo-test performance for RD-Agent candidates, but the reasons differ by phase. The raw candidates in Phases 3A-3C did not pass their respective scoring-admission protocols. In Phase 3D, it was the controller-normalized candidate that passed, and the research endpoint was limited solely to structural and data-boundary compliance. In Phase 3E, the preregistered endpoint was the structural-compliance rate of first-attempt raw candidates. The protocol explicitly prohibited using performance metrics to determine success and did not authorize structurally compliant candidates to continue accessing pseudo-test labels.

The three successful Phase 3E runs therefore indicate only that the candidates passed the structural-compliance gate. They do not indicate that pseudo-test scores were obtained or that predictive performance exceeded the human-controlled baseline. If pseudo-test scoring were temporarily opened after the structural audit, failed candidates were manually modified, or additional attempts were made, three types of methodological problems would arise:

- It would change the preregistered research endpoints, attempt limits, or stopping rules;
- It would mean that the final results no longer correspond to RD-Agent's first raw outputs; and
- It would introduce researcher degrees of freedom through repeatedly adjusting the process based on intermediate results.

The absence of pseudo-test scores is therefore the result of the combined effect of the study design and scoring-access controls, rather than missing files or the experiment having been overlooked and not carried out. Accordingly, this study answers the question of whether an autonomous agent can reliably produce executable, verifiable, and compliant candidates. Predictive performance must be evaluated in a separately preregistered follow-up experiment with an independent evaluation boundary.

10.9 Phase 3 Summary

Five progressive sub-experiments form a clear governance chain:

Sub-experiment	Agent permissions	Main failure types	Final status
Phase 3A	Generate full-model code	Non-TRAIN categorical vocabulary; prohibited ensembles	Blocked
Phase 3B	Full-model code on structured data	Incomplete output; interface and model-structure violations	Blocked

Sub-experiment	Agent permissions	Main failure types	Final status
Phase 3C	Generate only restricted model recipes	Recipe Schema and parameter-contract violations	No qualified candidates; terminated
Phase 3D	Normalize near-compliant recipes under a frozen controller	Six prohibited `epsilon` fields and interface-specification issues	Controller-normalized candidate passed structural audit
Phase 3E	Independently generate first-attempt raw candidates; no repairs permitted	Syntax, Schema, parameter, and output-location violations	3/15 candidates achieved structural compliance on the first attempt

As agent permissions gradually shrink from full modeling to restricted recipe generation, the obvious risk of data leakage is controlled, but protocol compliance is not automatically achieved. The experiments show that autonomous agents in high-risk financial modeling require external controllers, machine-executable contracts, immutable candidate freezing, and independent admission checks to work together to form a closed loop.

From a business perspective, this mechanism separates the question of whether a model is worth scoring from the question of what score the model receives. Only after a model has passed the former does the latter have interpretability and governance significance.

Phases 3D and 3E further show that whether a near-compliant recipe can be recovered by the controller and whether the agent can independently achieve compliance on the first attempt are two different issues. The former was established in deterministic recovery experiments; the observed rate of the latter was only 20.0% under this configuration. Therefore, for autonomous agents in high-risk financial modeling, the principal shortcoming is not only proposing a modeling approach, but also reliably converting the approach into executable, verifiable, and correctly placed machine artifacts.

11. Business Application Solutions

The model can serve as a quarterly credit risk screening tool rather than an automatic substitute for analyst judgment.

The recommended business process is as follows:

- Generate risk probabilities quarterly for all active issuers;
- Rank issuers by their risk probabilities to form a prioritized watchlist;
- Select a Top-K list or a risk threshold while taking risk capacity into account;
- Have analysts review rating changes, financial deterioration, and unusual missingness patterns;
- Record whether early warnings are adopted and the subsequent actual outcomes;
- Each quarter, monitor drift, ranking ability, and calibration; and
- Trigger retraining when performance or data quality crosses prespecified thresholds.

The model output is better suited to relative risk ranking than to direct interpretation as an uncalibrated absolute rating-downgrade probability. Business assessments should also focus on how many future rating-downgrade events can be captured with fixed review resources.

12. Main Conclusions

This study yields five main conclusions.

First, the foundation of a real-world financial forecasting project is not a single model fit, but the construction of an auditable point-in-time dataset. Financial-disclosure lags, multiple versions of the same reporting period, and rating-event windows must be strictly aligned at the prediction time point; otherwise, any subsequent performance results may be contaminated by future information.

Second, there are predictive signals for bond-rating downgrades that can be used for risk ranking, but PR-AUC must be interpreted in conjunction with event rates and actual review capacity. Although the human-controlled model's hidden-test PR-AUC fell from 0.2764 in the validation period to 0.0866, it remained 25.27 times the test event rate and captured one-third of downgrade events in the Top 1% list. The model therefore retains screening value, but outputs that have not been recalibrated should not be directly interpreted as absolute risk probabilities.

Third, model reliability has a clear time dependence. Rolling backtests, quarterly metrics, issuer-level Cluster Bootstrap, and PSI analyses jointly show that event rates, input distributions, and ranking performance all change over time. A single validation set cannot represent deployment performance in more distant future periods, and model monitoring needs to simultaneously cover performance, probability scale, data coverage rate, and drift in representative features.

Fourth, progressively restricting RD-Agent permissions can reduce leakage risk, but cannot automatically guarantee protocol compliance. Phase 3D shows that a frozen controller can recover a near-compliant recipe without using performance feedback. Phase 3E shows that only 3 of 15 first-attempt raw candidates passed the structural-compliance gate without controller repair, yielding an observed pass rate of 20.0% (Wilson 95% confidence interval, 7.05%-45.19%). Therefore, whether the controller can recover and whether the agent can independently achieve compliance on the first attempt are two distinct reliability propositions. Both natural-language instructions and machine-readable interfaces require independent execution-layer verification.

Fifth, reliable autonomous data science requires closed-loop governance:

- Data layer: uses time-consistent joins, TRAIN-only preprocessing, and physical isolation of high-risk fields;
- Experimental layer: predefines time windows, metrics, the number of candidates, and model-selection rules;
- Execution layer: assigns training, prediction alignment, and scoring logic to the external controller;
- Admission layer: freezes candidates and conducts independent protocol checks; and
- Evaluation layer: allows only compliant candidates to access hidden labels or undergo pseudo-test scoring.

Therefore, RD-Agent is more suitable as a controlled research component in a corporate risk modeling environment rather than a decision-making system operating without human supervision and directly holding permissions for data access, training, selection, and scoring.

13. Limitations and Follow-Up Work

This study has the following limitations:

- The original hidden test set was used for one formal scoring of the sole protocol-compliant human-controlled candidate only; therefore, it cannot be used for subsequent model selection or repeated parameter tuning;

- Noncompliant RD-Agent candidates have neither hidden-test nor pseudo-test scores and therefore cannot be ranked directly by performance; this absence results from preregistered admission rules rather than insufficient logs from candidate runs;
- The issuer-level cluster bootstrap holds the model and predictions fixed, without repeating training, feature engineering, or candidate selection; it therefore reflects only conditional uncertainty caused by changes in test-sample composition;
- PSI measures marginal input distribution changes and cannot prove changes in feature-label relationships, nor can it establish causal attribution for PR-AUC decay;
- Fold 3 overlapped with the original hidden-test period and was not used for model comparison during development;
- PR-AUC and Top-K capture rate mainly measure ranking and screening ability and are not equivalent to direct economic value;
- Business-optimal thresholds have not yet been estimated with real analyst review costs, losses from missed downgrade events, and risk capacity;
- A systematic stratified evaluation by industry, rating grade, rating agency, and new versus existing issuers has not yet been completed; and
- Phase 3E includes only 15 primary-evaluation samples. Although the observed first-attempt structural-compliance rate is 20.0%, its Wilson 95% confidence interval is 7.05%-45.19%. This result applies only to this task, MiniMax-M3, temperature 0.5, and the locked execution environment. It must not be generalized directly to other agents, model versions, prompt configurations, or execution environments, and it cannot be used to estimate the magnitude of improvement from stricter generation-time validation.

Follow-up research should establish new prospective observation windows without compromising the existing hidden-test boundary, while locking development and evaluation plans in advance. Further research may include:

- Discrete-time hazard models and survival analysis to explicitly express event timing and censoring;
- Multi-task learning and time-varying risk modeling with different early-warning horizons;
- Probabilistic calibration and calibration-drift evaluation in the new development window;
- Decision-curve and threshold analyses that incorporate review costs and risk capacity;
- Stability assessments stratified by industry, rating grade, rating agency, and new issuers;
- Consistency and temporal stability of SHAP explanations and case-level interpretations;
- Uncertainty evaluation for the full process under issuer-level resampling;
- Automated monitoring of data quality, PSI, unknown categories and risk list stability;
- Direct enforcement of Schema validation and type constraints at the agent-output stage so that invalid recipes receive machine feedback before candidate freezing; and
- A least-privilege closed loop among the external controller, independent auditor, and RD-Agent.

Conclusion

This project integrates publicly available market data, point-in-time data engineering, rare-event prediction, rolling temporal validation, uncertainty analysis, distribution-drift diagnosis, and autonomous-agent governance into an auditable workflow.

The findings of this study demonstrate that early warnings of bond-rating downgrades can provide practical ranking value in an extremely low-event-rate setting, while also showing that model performance can change substantially with market conditions, data coverage, and issuer composition. For RD-Agent, the focus of the experiment is not to pursue an unconstrained maximum score, but to test whether candidates can qualify for scoring within prespecified data and methodological boundaries.

Therefore, what this study ultimately produces is not a single model result, but a governance framework spanning time-consistent data construction, candidate generation, independent auditing, hidden evaluation, and deployment monitoring. The framework provides a reusable methodological foundation for the company's future credit-risk modeling, automated experiment management, and model-admission processes.